

● EASY

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

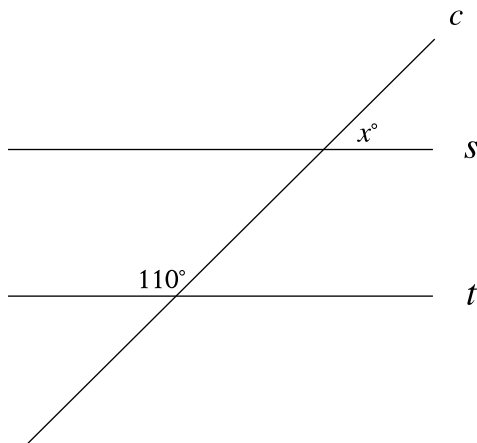
07

10 questions · ~15 min

Q1

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled, c , s , and t .
- Line c intersects both line s and line t .
- At the intersection of line c and line s , 1 angle is labeled clockwise from top left as follows:
 - Top right: x°
- At the intersection of line c and line t , 1 angle is labeled clockwise from top left as follows:
 - Top left: 110°
- A note indicates the figure is not drawn to scale.

In the figure shown, line c intersects parallel lines s and t . What is the value of x ?

STUDENT-PRODUCED RESPONSE

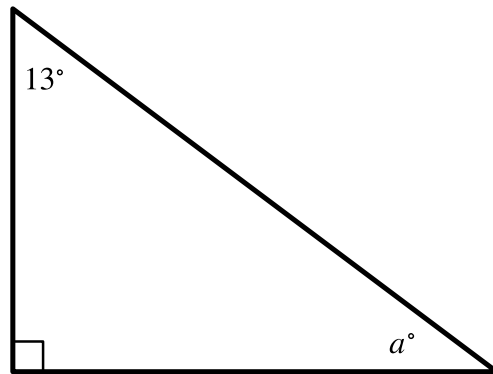
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The area of a rectangle is 630 square inches. The length of the rectangle is 70 inches. What is the width, in inches, of this rectangle?

- A. 9
- B. 70
- C. 315
- D. 560



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measures of the other 2 angles are as follows:
 - a°
 - 13°
- A note indicates the figure is not drawn to scale.

In the right triangle shown, what is the value of a ?

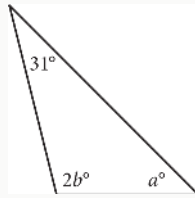
- A. 13
- B. 77
- C. 90
- D. 103

Q4

GEOMETRY AND TRIGONOMETRY

The area of a square is 64 square inches. What is the side length, in inches, of this square?

- A. 8
- B. 16
- C. 64
- D. 128



In the triangle above, $a = 45$. What is the value of b ?

- A. 52
- B. 59
- C. 76
- D. 104

Q6

GEOMETRY AND TRIGONOMETRY

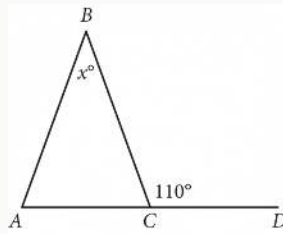
What is the area, in square inches, of a rectangle with a length of 7 inches and a width of 6 inches?

A. 13

B. 20

C. 42

D. 84



In the given figure, $\overline{AC}$ extends to point D. If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of x ?

- A. 110
- B. 70
- C. 55
- D. 40

Each base of a right rectangular prism has a length of 19 inches and a width of 8 inches. The prism has a volume of 2,736 cubic inches. What is the height, in inches, of the prism?

- A. 18
- B. 27
- C. 144
- D. 152

In $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . What is the measure of $\angle Z$?

- A. 43°
- B. 89°
- C. 91°
- D. 179°

The table gives the perimeters of similar triangles TUV and XYZ , where TU corresponds to XY . The length of TU is 6.

	Perimeter
Triangle TUV	50
Triangle XYZ	150

What is the length of XY ?

- A. 2
- B. 6
- C. 18
- D. 56